

Problem Set-1

Problem1. This elementary problem begins to explore propagation delay and transmission delay, two central concepts in data networking. Consider two hosts, A and B, connected by a single link of rate R bps. Suppose that the two hosts are separated by m meters, and suppose the propagation speed along the link is s meters/sec. Host A is to send a packet of size L bits to Host B.

- a.) Express the propagation delay, d_{prop} , in terms of m and s .
- b.) Determine the transmission time of the packet, d_{trans} , in terms of L and R .
- c.) Ignoring processing and queuing delays, obtain an expression for the end-to-end delay.
- d.) Suppose Host A begins to transmit the packet at time $t = 0$. At time $t = d_{\text{trans}}$, where is the last bit of the packet?
- e.) Suppose d_{prop} is greater than d_{trans} . At time $t = d_{\text{trans}}$, where is the first bit of the packet?
- f.) Suppose d_{prop} is less than d_{trans} . At time $t = d_{\text{trans}}$, where is the first bit of the packet?
- g.) Suppose $s = 2.5 * 10^8$, $L = 120$ bits, and $R = 56$ kbps. Find the distance m so that d_{prop} equals d_{trans} .

Problem2. Suppose users share a 3 Mbps link. Also suppose each user requires 300 kbps when transmitting, but each user transmits only 10 percent of the time.

- a.) When circuit switching is used, how many users can be supported?
- b.) For the remainder of this problem, suppose packet switching is used. Find the probability that a given user is transmitting.
- c.) Suppose there are 20 users. Find the probability that at any given time; exactly n users are transmitting simultaneously. (Hint: Use the binomial distribution.)
- d.) Find the probability that there are 18 or more users transmitting simultaneously
- e.) Find the probability that there are 4 or more users transmitting simultaneously

Problem3. Calculate the latency (from first bit sent to last bit received) for the following:

(a) 100-Mbps Ethernet with a single store-and-forward switch in the path and a packet size of 12,000 bits. Assume that each link introduces a propagation delay of $10\ \mu\text{s}$ and that the switch begins retransmitting immediately after it has finished receiving the packet.

(b) Same as (a) but with three switches.

(c) Same as (a), but assume the switch implements “cut-through” switching; it is able to begin retransmitting the packet after the first 200 bits have been received.